

SKELETON-BASED ANOMALY DETECTION IN CITATION NETWORKS

25-2-R-2

MOTIVATION



Dynamic Nature

Citation networks evolve constantly, static models fail to capture this dynamics.



Complexity

Hierarchical structures and complex behaviors require continuous-time modeling.



Challenge

Existing methods miss temporal evolution.

PROBLEM RESOLUTION



Goal

Predict future citation behaviors and detect structural anomalies.



Method

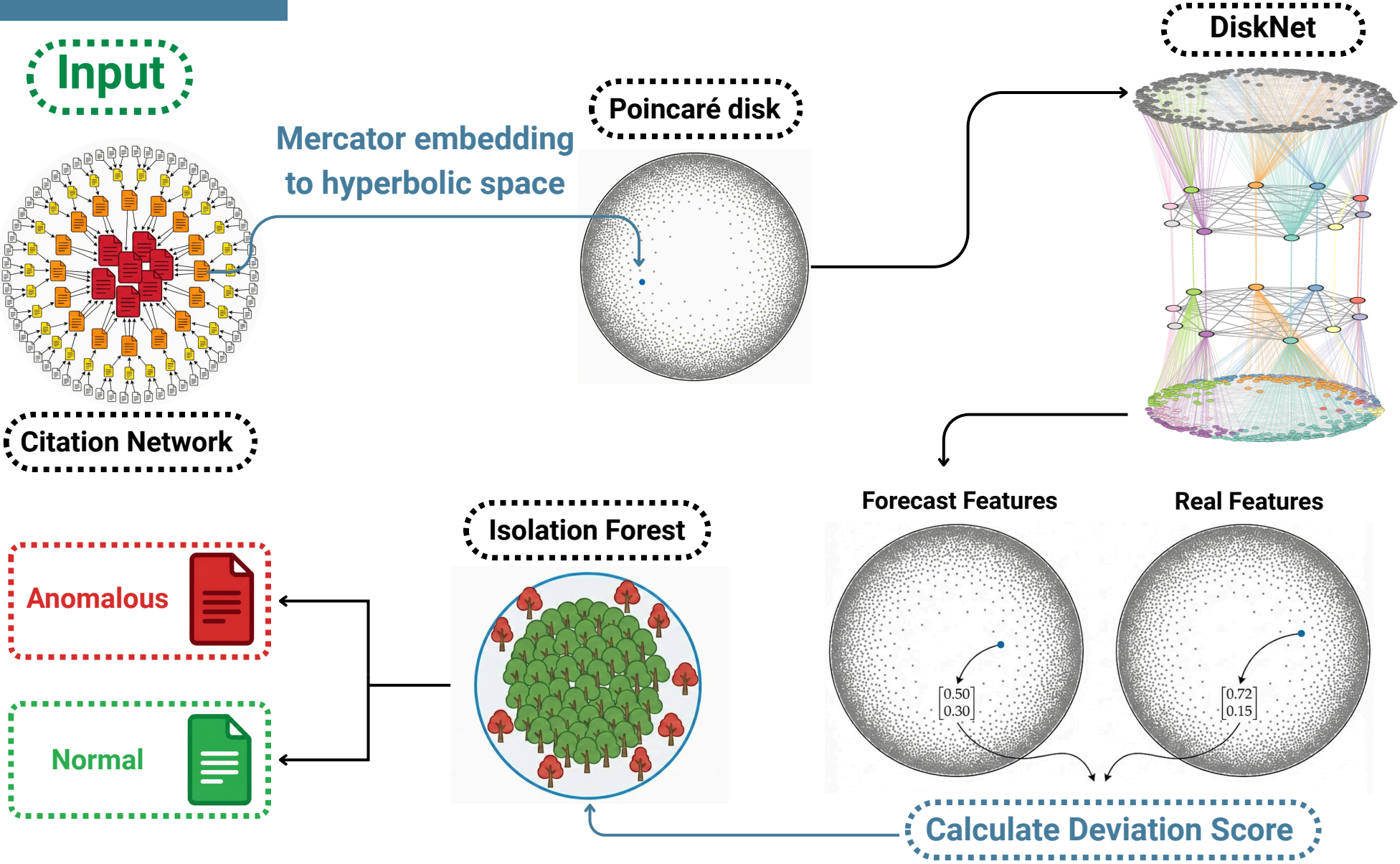
A geometry-aware framework combining Hyperbolic Embeddings with Neural ODEs.



Detection

Identify nodes deviating from their predicted evolutionary trajectory.

WORKFLOW



DISKNET

Hyperbolic Geometry

Utilizes the Poincaré disk model, as its negative curvature naturally accommodates the exponential growth and hierarchical topology of citation networks.

Skeleton Extraction

Aggregating geometrically similar nodes into "Supernodes" to reduce dimensionality and noise.

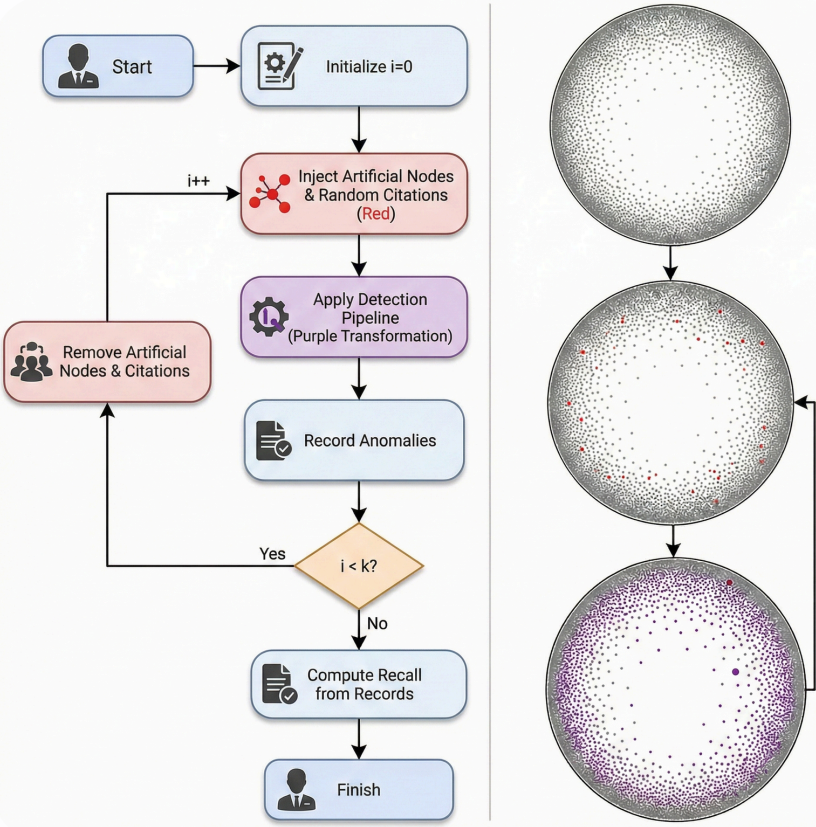
Neural ODEs

Modeling the continuous-time evolution of the "skeleton" in the latent space.

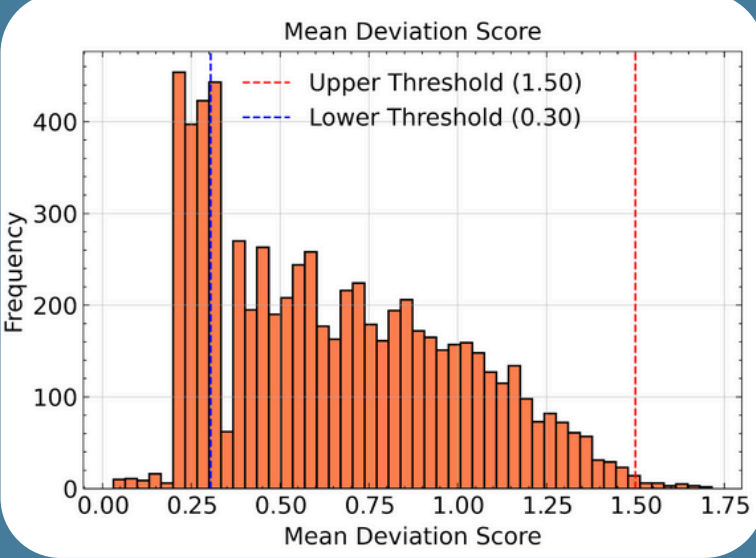
Lifting & Refinement

Projecting coarse skeleton dynamics back to fine-grained node trajectories for anomaly scoring.

VALIDATION PROCESSES



RESULTS



The Figure reveals the separation between normal and anomalous nodes. The legitimate papers are clumped between the thresholds



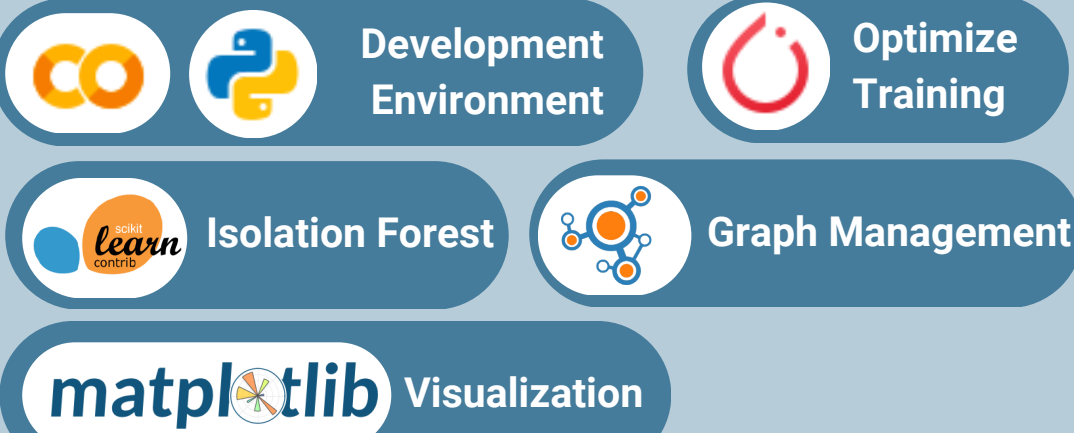
Hyperbolic embedding

A dense core of standard papers contrasts with distinct peripheral anomalies.

Detection Results	Avg (10 Iterations)
Original Anomalies	1020
Original Anomalies (%)	16.32
Synthetic Anomalies	453
Synthetic Anomalies (%)	72.74

Consistency: The model demonstrated exceptional stability across 10 independent iterations, achieving a consistent average Recall of ~72.7%.

IMPLEMENTATION TECHNOLOGIES



CONCLUSIONS

- ✓ Citation features are a primary driver of embedding performance.
- ✓ Provides a scalable solution for maintaining academic database integrity.
- ✓ Structural anomalies are detectable via latent hyperbolic geometry deviations.